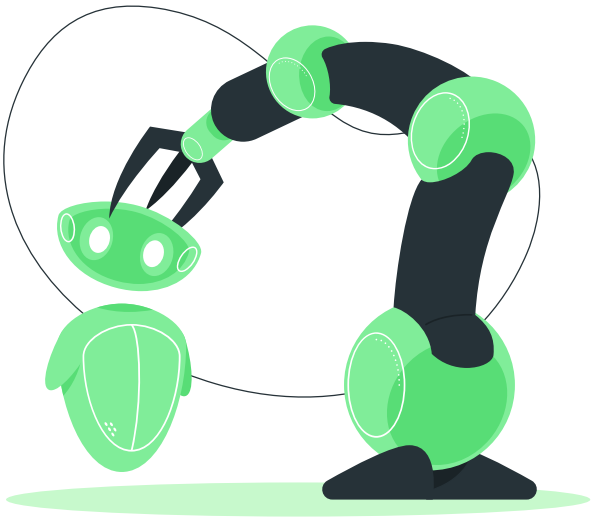
Robotics
CS 3015

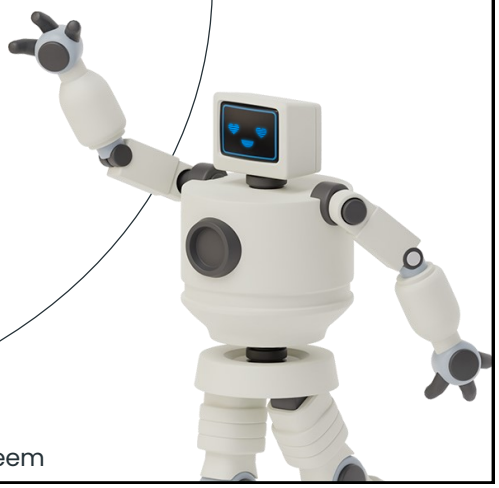
Umarah Qaseem



1

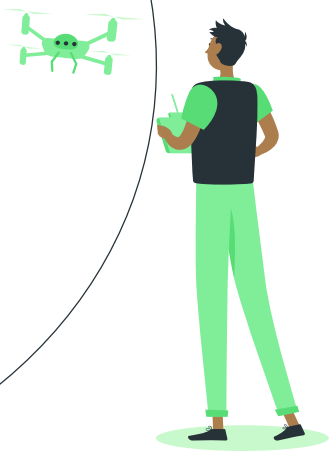
15

- Sensors Types
- based on output
- Based on location
- Based on application
- Based on Environment
- Shaft Encoders
- Incremental, Absolute
- Sonar
- Ultra Sonic
- LIDAR
- LIDAR Scans
- LASER
- IR
- Accelerometers
- Gyro



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Sensors



Robotics – Umarah Qaseem

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Sensors Types
Based on Sensor Output

- Binary signal (0 or 1)
  - Door sensor, Tactile Sensor
- Analog signal (e.g., 0 … 5 V)
  - Temperature Sensor
- Timing signal (e.g., PWM)
  - Proximity Sensor
- Serial link (RS232, SPI, I2C)
  - GPS
- Parallel link
  - Camera

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Sensors Types
Based on location

- Local sensors
  - sensors mounted on the robot
- Global sensors
  - sensors mounted outside the robot in its environment and transmitting sensor data back to the robot.

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Sensors Types
Based on Application

- Internal (proprioceptive) sensors
  - sensors monitoring the robot's internal state
- External sensors
  - sensors monitoring the robot's environment.

7

Sensors Types
Based on Effect on Environment

- Passive sensors
  - sensors that monitor the environment without disturbing it, for example digital camera, gyroscope
- Active sensors
  - sensors that stimulate the environment for their measurement, for example sonar sensor, Lidar laser scanner, infrared sensor.

8

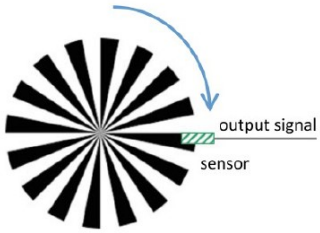
Shaft encoders

- Shaft encoders are standard sensors on mobile robots for determining their position, speed and orientation.
- Read the position of a rotating disk
- Converts the angular position or motion of a shaft or axle to analog or digital output signals
  - Magnetic
  - Optical

9

Shaft encoders

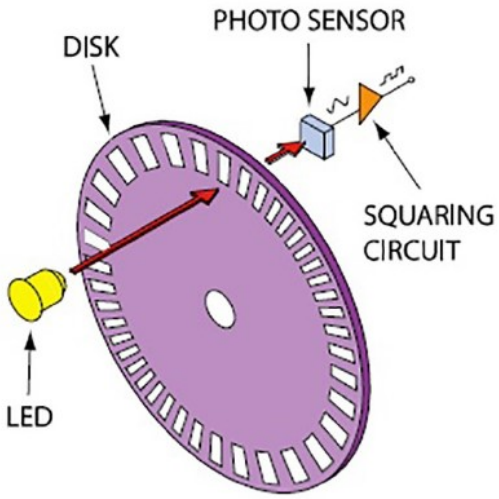
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10

Shaft encoders

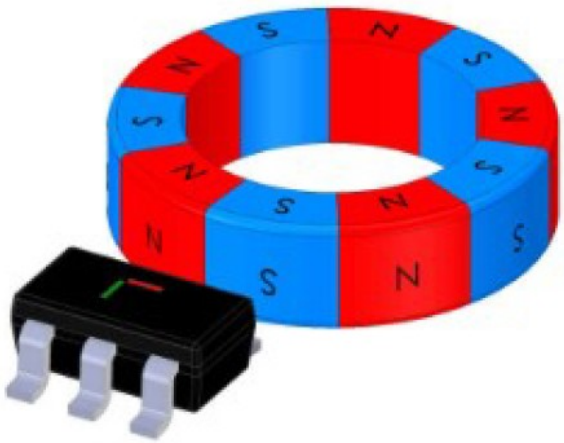
- Optical



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Shaft encoders

- Magnetic



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Shaft encoders

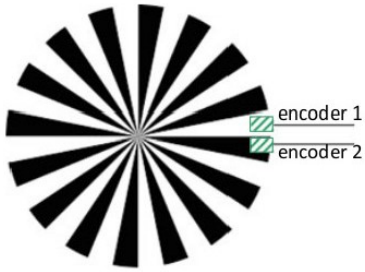
Signal for encoder disk spinning

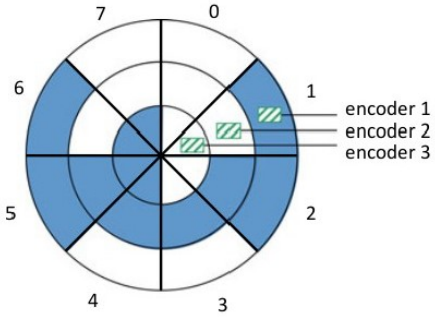
- Slow:
- Fast:
- Not at all:

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Shaft encoders

- Incremental
- Absolute





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Incremental Shaft Encoder

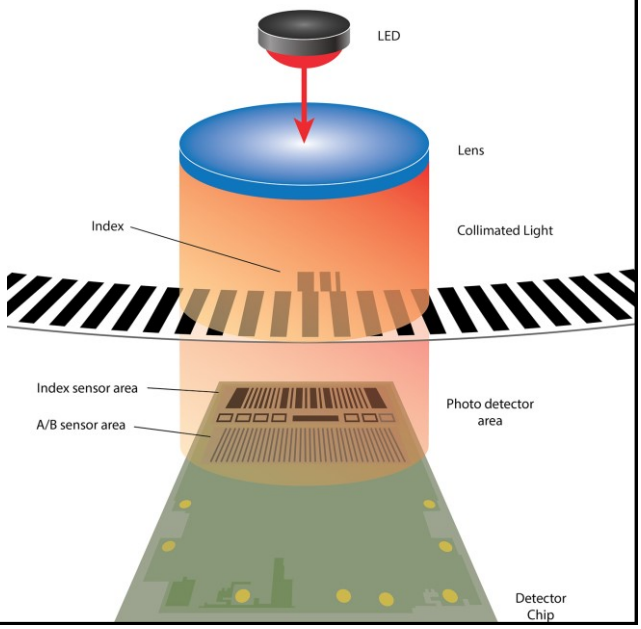
- Simple and very cost effective
- Give relative/incremental position
- How about rotation direction?



15

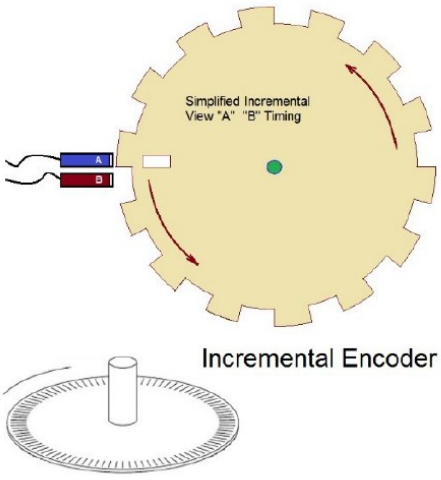
Incremental Shaft Encoder

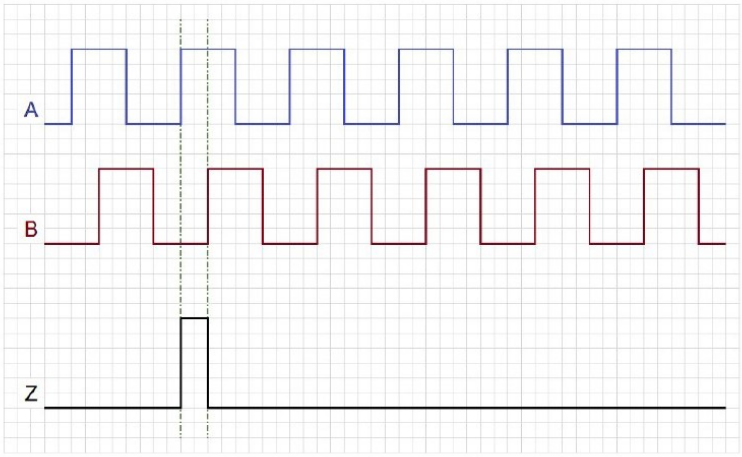
- Simple and very cost effective
- Give relative/incremental position
- How about rotation direction?



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Incremental Shaft Encoder



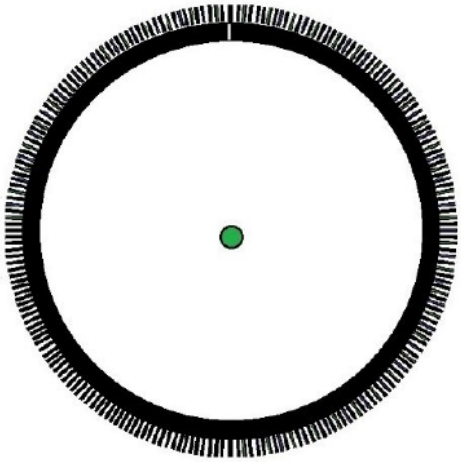


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Incremental Shaft Encoder

- Simple and very cost effective
- Give relative/incremental position
- How about rotation direction?

- For Animation:
- https://www.youtube.com/watch?v=zzHcsJDV3_o

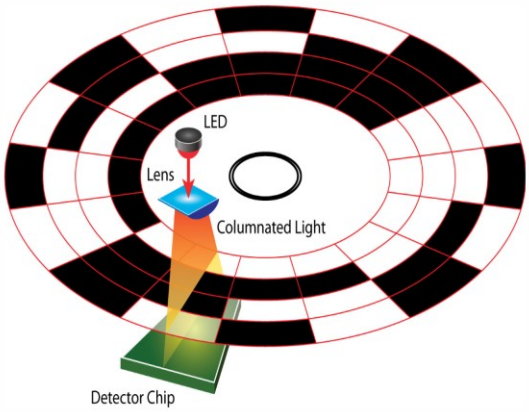


Incremental Encoder

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Absolute Shaft Encoder

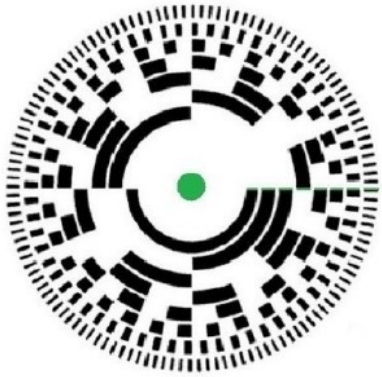
- Provide a direct bit representation position by “reading” the Binary value associated its current position.



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Absolute Shaft Encoder

- Provide a direct bit representation position by “reading” the Binary value associated its current position.

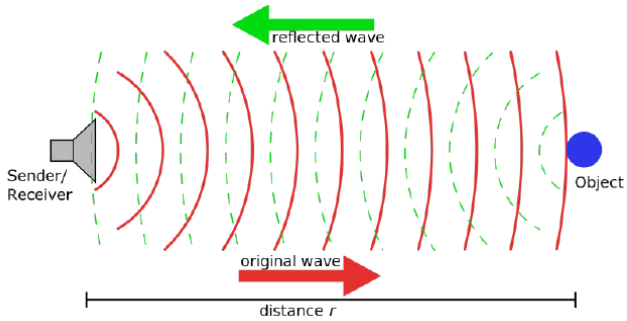


8 Bit Binary

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Sonar (sound/sonic navigation and ranging)

- uses sound propagation to navigate, measure distances
- usually underwater (submarine navigation)
- Detect objects on or under the surface of the water, such as other vessels
- Problems:
  - Inaccuracy
  - Reflections
  - Interference



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Ultra Sonic Sensor

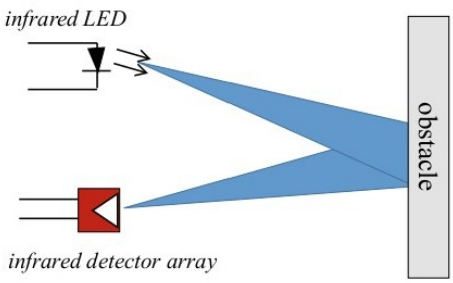
- A mini version of Sonars
- Basic functionality principle is same.



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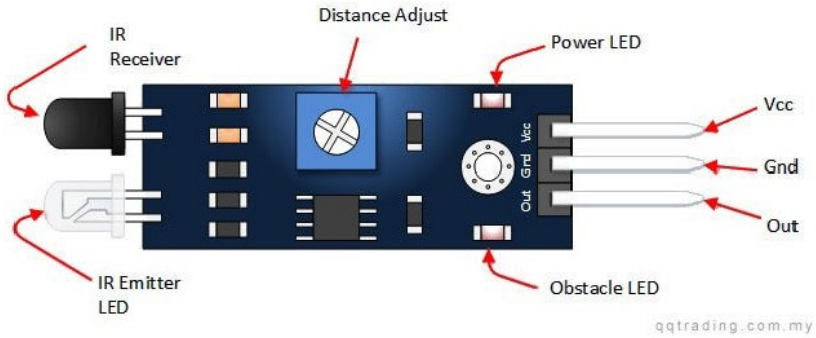
IR Sensor

- It uses infrared light.
- The angle under which the reflected beam is received, changes according to the distance to the object and therefore, can be used as a measure of the distance.



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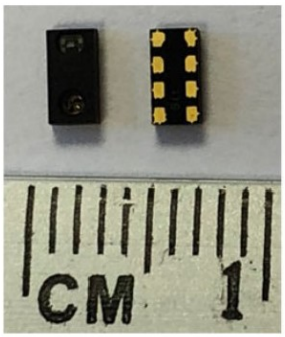
IR Sensor for arduino

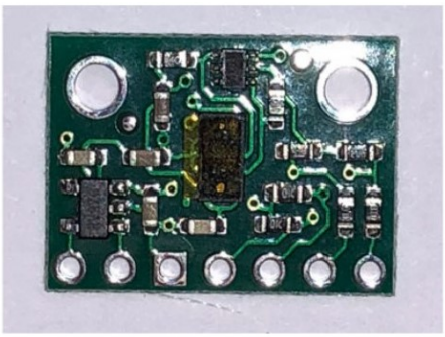


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LASER Sensor

- Inexpensive monochromatic single-beam time-of-flight measurement
- Connected to controller on I2C interface
- It uses focused laser beam





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LIDAR

- Light Detection and Ranging
- One or more laser sources
- Automotive LIDARS: 8, 16, 32 beams
- LIDAR distance data is simpler to process as compared to CAMERA
- Expensive
    - Requires high performance timing sensors
    - Due to ToF measurement of light

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LIDAR Animations

- LIDAR Working
- https://youtube.com/shorts/T_6c5sGQGIQ?si=bXde64zw_NfVAjfn

- LIDAR VS RADAR
- https://youtube.com/shorts/7T4EuZoYk6s?si=sjPwLuKY_cI_GWqe

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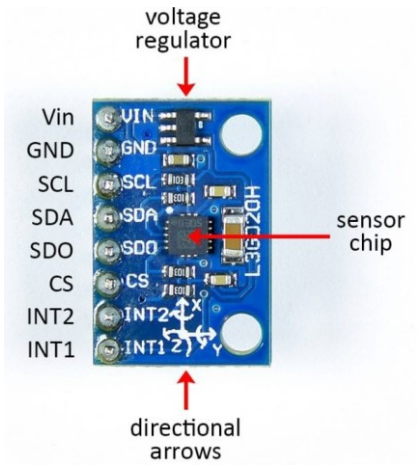
Accelerometers

- Accelerometers measure acceleration
- Rate of change in speed
- Can be single or multiple axis
- accelerometers can detect acceleration and convert it
- into measurable quantities – like electrical signals
- Variety of applications, especially in robots

- Covered in detail along with magnetometers and Gyroscope.

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L3GD20H gyro



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Credits

This lecture is adapted from Dr Imran Ashraf's lecture content on the topic of sensors.

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